



ILLUSTRATION BY RAJ VERMA

THE BURNING ISSUE

There have been incidents of scooters from newer EV players catching fire. What's the problem and how do you solve it?

By **Chetan Bhutani**

ELECTRIC VEHICLES are on fire—figuratively and, sadly, literally too. Sales of electric two-wheelers have gone through the roof, with the Federation of Automobile Dealers Associations reporting a rise of over 450 per cent over the previous year, even though volumes remain low. However, viral videos of electric two-wheelers catching fire have the potential to bring a cloud over consumers' enthusiasm.

Electric scooters have been around for a while, but

such incidents haven't been reported earlier. According to Sohinder Gill, CEO of Hero Electric and Director General of the Society of Manufacturers of Electric Vehicles, one reason for such fires could be that some newer players have rushed their vehicles to the market without spending proper time on R&D. Established automotive players, he says, take time in bringing EVs to the market and don't do "R&D on the road but in their backyard". In a chat with *Business Today* TV, Gill says: "There is certainly a need, especially on the battery side, of simulating the end-of-life conditions because deterioration can happen any time between 0-7 years... Simulation techniques, testing techniques are available and they have to be deployed properly on a large base."

Akshay Singhal, Founder & CEO of battery technology firm Log9 Materials, says one cannot apply a software mindset to a hardware play, adding that there is a need for indigenised, locally developed technologies: "It's not like you can bring any technology from outside and force fit it in the Indian context. You have to be mindful of our requirements and climatic conditions."

Experts say these fires are primarily thought to be caused by overheating, short circuits and extreme vibration or stress on the lithium-ion battery units. Lithium, cobalt and nickel are some of the raw materials used to manufacture these batteries—and cobalt is highly flammable. The key to tackling such issues are the Battery Management System (BMS) algorithms, which ensure a battery's safety and efficiency. Rajiv K. Vij, Chairman & MD of EV mobility-as-a-service platform Plug Mobility, says such incidents take place because the OEMs have not used appropriate technology for BMS/battery cooling and/or ignored quality standards.

There are other issues, too. Sushant Kumar, Founder & MD at electric two-wheeler maker AMO Mobility Solutions, suggests that OEMs should maintain a fine balance between the speed and range of the vehicle. Second, customers need to be educated on how to maintain their EV, especially the battery. "One should always use genuine accessories... [and] control the temptation of fast charging [as]... this area needs R&D," he says. Third, people also need to consider the external environment while using EVs. "While charging the vehicle, we should make sure that the outside temperature is not too hot and there is proper ventilation," he adds.

Hopefully, these incidents will not dampen the enthusiasm of customers. Gill says the sale of EVs could hit one million by the end of FY23, adding that industry players should take a step back now. "They can still roll back what they were doing and re-introduce a product [after proper testing]," he says. For electric scooters, it seems that the slow and steady will win the race. **BT**

@bhutanichetan